

Effect of EOG Artifact Removal on EEG Motor-Imagery Classification

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Abstract—Deep Learning (DL) models have proven to be efficient in Motor Imagery Brain Computer Interface (MI-BCI) task classification from non invasive Electroencephalogram (EEG) signals. However, most neural network models demand high computational power and vast memory to attain satisfactory performance. This poses a major challenge to the edge level implementation of EEG based motor imagery classification. In this paper, the computation and memory limitations of an edge level devices is considered and a compact and temporally aware neural network model with an efficient data pre-processing step is introduced. The model uses a light-weight linear regression model for EOG artifact removal and are parallel-placed Temporal Convolutional Neural Network that can cover small receptive fields along with a conventional exponentially increasing receptive field coverage. Through the model, 4-class MI task classification is performed on the "BCI Competition 2008 - data set 2A", scoring 80.5% in mean accuracy for all 9 subjects, with a standard deviation of 9.54%. For the reduced standard deviation, a single neural network structure with fixed hyper-parameter will be sufficient for resource-limited devices at the edge.

Index Terms—BCI, EEG, MI task Classification, CNN

I. INTRODUCTION

The process of enabling direct communication among humans and external devices, through the analysis of human brain neural activity is called Brain-Computer Interfaces (BCI) [1]. For BCI, non-invasive electroencephalography (EEG) signals are usually used. In order to manifest BCI, Motor-Imagery (MI) or the cognitive process of thinking of movement of body parts, prove to be a promising approach. But, due to MI-EEG signals having low signal to noise ratio (SNR) and high subjective variance, interfacing is difficult. Conventional approaches include Filter Bank Common Spatial Patterns (FBCSP) [2], Reimannian co-variance [3] and other handcrafted feature extractors coupled with classifiers such as Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM). These approaches heavily rely on domain-knowledge.

Recently, the Convolution Neural Network (CNN) based approaches are gaining traction, reducing dependence on domain-knowledge and hand-crafted feature extractors. Models such as Shallow ConvNet [4], TPCT [5], EEGNet [6] are being used for MI-task classification [7]. The current SOA model for Motor Imagery Task classification is held by TPCT [5], which has a 88.87% accuracy on 4-class MI BCI Competition 4-2A Dataset [8].

Conventional MI classifier neural network models are deployed on desktop platforms or cloud platforms that have con-

cerns with latency, availability and privacy. Moreover, using single MI-BCI model dose not perform well for all subjects [9]. Hence, training the models in Edge level devices such as low power micro-controllers is adopted [10]. This limits the heavier models such as the TPCT (1.73 billion trainable parameters), and compact models like the EEGNet, with 2.5k trainable parameters are preferred and proven useful in MCU implementations [11], [12]. EEGNet poses the drawback of high standard deviation and low accuracy of 72.4%. Another model, EEG-TCNet [13] implements a temporal Convolution Neural Network module, that exploits the temporal features extracted from EEGNet and performs classification with 77.6% accuracy and 12% standard deviation. Hence, a model with lower standard deviation, preserving accuracy performance is highly desirable.

In this paper, we introduce a pre-processing step of EOG artifact removal, in addition to the conventional pre-processing steps, and also introduce a Neural network model inspired from EEG-TCNet, in addition, includes an extra TCN [14] block that caters to low receptive field features, which lead to an experimentally obtained lower standard deviation and higher mean accuracy than the SOA model EEG-TCNet.

The main contribution of the paper are as follows:

- A neural network model inspired from EEG-TCNet with 80.5% mean accuracy, surpassing EEG-TCNet by 4% mean accuracy, and standard deviation of 9% improving the standard deviation of EEG-TCNet by 3%.
- EOG Artifacts removal through linear regression. The impact of the pre-processing step is apparent in reproduction of both EEGNet and EEG-TCNet.
- Evaluation of our model and pre-processing based approach with respect to existing benchmarks [15] of neural network models on the BCI Competition IV-2a Data-set.

The codes have been made available online under permissible open-source license.

II. METHODOLOGY

A. Dataset

BCI Competition IV-2a dataset has been used for benchmarking. This data set contains EEG data from nine different people. The four motor imagery tasks in the cue-based BCI paradigm were imagination of movement of the left hand (class 1), right hand (class 2), both feet (class 3), and tongue

(class 4). For each subject, two sessions were recorded on different days. Each session consists of six runs with short pauses in between. A session consists of 288 trials, with 48 trials each run (12 for each of the four potential classes). A recording of around 5 minutes was made at the start of each session to estimate the EOG influence. The recording was split into three sections: (1) two minutes of open eyes (looking at a fixation cross on the screen), (2) one minute of closed eyes, and (3) one minute of eye movements.

The participants were seated in an armchair before a computer screen. A fixation cross displayed on the dark screen at the start of each trial ($t = 0s$). Furthermore, a short alarm tone was played. A cue in the form of an arrow pointing left, right, down, or up (corresponding to one of the four classes left hand, right hand, foot, or tongue) came after two seconds ($t = 2s$) and stayed on the screen for 1.25 seconds. The subjects were then encouraged to complete the necessary motor imagery task. There was no response. The participants were instructed to complete the motor imagery task until the fixation cross vanished from the screen at $t = 6$ seconds.

The EEG was recorded using twenty-two Ag/AgCl electrodes (with inter-electrode spacing of 3.5 cm). All signals were recorded in a monopolar method, with the left mastoid serving as the reference and the right mastoid serving as the ground. The data was captured at 250 Hz and band pass filtered between 0.5 and 100 Hz. The amplifier's sensitivity was set to 100 volts. To reduce line noise, an additional 50 Hz notch filter was imposed.

Three monopolar EOG channels were captured and sampled with 250 Hz in addition to the 22 EEG channels. They were band pass filtered from 0.5 to 100 Hz (with the 50 Hz notch filter enabled) and the amplifier's sensitivity was adjusted at 1mV. The EOG channels are intended to be used in conjunction with artifact processing methods, not for classification. An expert performed a visual review of all data sets, and trials with artifacts were flagged. A total of eight of the nine data sets were examined.

B. EOG removal through Linear Regression

In agreement with the work of [16], we assume the following linear model with three spatial (horizontal, vertical, and radial) EOG components:

$$Y(t, ch) = S(t, ch) + [EOG1(t), EOG2(t), EOG3(t)] * [b_1(ch), b_2(ch), b_3(ch)]^T \quad (1)$$

where, $Y(t, ch)$ is the recorded value of channel ch at time t , S is the source signal without artifact contamination, $EOG123$ indicates the noise source, U , of the three spatial EOG components, and $b(ch)$ indicates the weights of the EOG artifacts at the EEG channel ch and T means matrix transposed. Extending this to more EEG channels and using a matrix notation, we can write:

$$Y_{T \times M} = S_{T \times M} + N_{T \times n} * b_{n \times M} \quad (2)$$

The indices indicate the size of each matrix; the signals Y and S have T time points and M channels, the noise, U ,

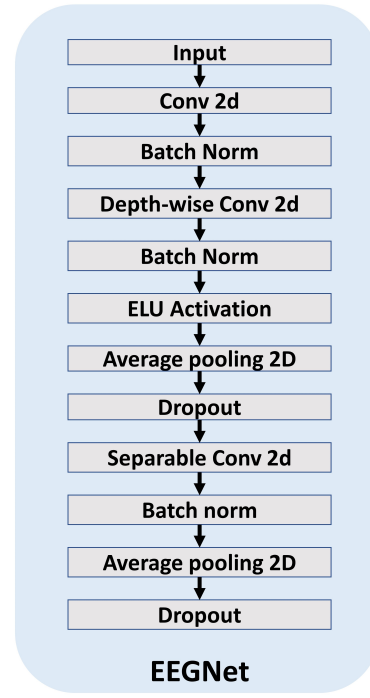


Fig. 1. EEGNet Module for Feature Extraction

has n components and b denotes the weights from each EOG component to each EEG channel; subsequently these indices are omitted. In order to obtain the corrected signal $S = Y - U * b$, the noise source, U , and its weighting coefficients, b , must be known. In this work, the noise source (i.e., EOG) was recorded by separate EOG channels. In order to identify the weighting coefficients, b , we assume the signal, S (i.e., EEG), and the noise, U (i.e., EOG), are independent, then

$$\langle U^T S \rangle = \langle U^T Y \rangle - \langle U^T U \rangle b \quad (3)$$

with $\langle U^T S \rangle = 0$ results in

$$b = \langle U^T U \rangle^{-1} \langle U^T Y \rangle = C_{NN}^{-1} C_{NY} \quad (4)$$

Here, $C_{NN} = \langle U^T U \rangle$ is the auto-covariance matrix of the EOG channels and $C_{NY} = \langle U^T Y \rangle$ is the cross-covariance between the EEG and EOG channels. Accordingly, the EEG can be corrected by the following equation:

$$S = Y - U * b \quad (5)$$

In order to obtain data for calculating the correction coefficients, a short data segment was recorded during which the subjects performed eye movements.

C. Temporal Feature Extraction

EEGNet inspired module is used for temporal feature extraction as the module is compact and has low trainable parameters. The possibility of parallel branches of EEGNet feature extractor was considered but was experimentally proven inferior to a single branch. The network structure is presented in Fig. 1.

Our model is highly motivated by EEGNet architecture, which learns a very detailed frequency filters from a 2D convolution followed by a depth-wise convolution. Batch normalization along with ELU activation function have been used to make the network more stable, fast, and accurate with lower computational complexity. Separable convolution blends the gist of each feature map. A 2D adaptive average pooling is used to reduce size of feature map.

Our network use two different TCN blocks which exact the temporal information. Then, we passed these to dense layer and predict the PDF for our desired classification using SoftMax [17] Activation.

D. Task Classifier

TCN (Temporal Convolution Network) module was used to perform task classification, exploiting the temporal feature extracted from EEGNet. Temporal Convolutional Networks (TCN) is a variation over CNN, but it differs in terms of causality, increase in receptive field and capacity of processing 2D feature maps using 1D convolutions. TCN uses Causal Convolutions to make sure that there is no leakage from the future, rather elements only from present time and past contributes. 1D FCN architecture is used to produce same length input output. Casual convolutions increase receptive field size linearly, so TCN uses sequence of dilated convolutions to produce exponential boost with ability to cover full input sequence using dilation factor, d . TCN also stacks two dilated convolutional layers. As a result, TCN have additional advantages like parallelism, flexible receptive field size, better adaptability, and low memory size, and so on. We have experimentally performed an analysis and have seen that two parallel branch of TCN blocks with different dilation factor prove to be very useful to capture small and largest values together (see Fig. 2 and Table I).

E. Training Procedure

On an NVIDIA GTX 1080 Ti GPU, models were trained and evaluated in a Tensorflow environment. Keras was used to create the networks. When training the models provided in this study, we employed this training configuration on a Google Collaboratory environment, where categorical cross-entropy loss is applied and the filter kernels are uniformly initialized. The models are trained using an Adam optimizer with a learning rate of 0.001 and a batch size of 64 for 750 epochs. Cross-validation on the training set is used to determine the training hyperparameters.

III. RESULTS AND DISCUSSIONS

A. Evaluation Matrices

The model was compared based on the classification accuracy, which represents the ratio of correct classification to total classification attempts in a test set.

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Predictions} \quad (6)$$

Standard deviation across different subjects in the dataset was observed to assess the feasibility of using a single

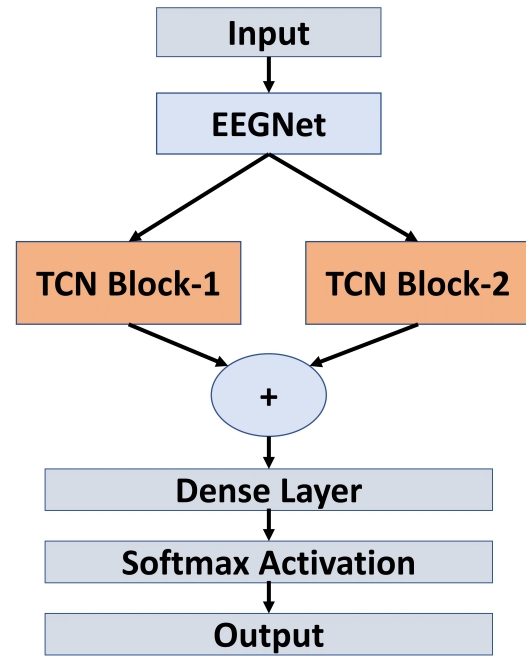


Fig. 2. Complete model Diagram

neural network structure with fixed hyper-parameter for task-classification.

StandardDeviation =

$$\sqrt{\frac{(Mean\ Accuracy - Subject\ Accuracy)^2}{Total\ Number\ of\ Subjects}} \quad (7)$$

The number of trainable parameters and memory size of the weights of the model were used to assess the feasibility of implementing the model in an edge level device with resource constrains.

B. Performance on different Subject

Concerned dataset consists of 9 subjects. The performance of a single fixed model architecture across the different subjects conventionally varies by greater than 10% standard deviation. Improved and consistent mean accuracy of the proposed model, compared to conventional architectures is presented in Table II, Fig. 3 and Fig. 4.

C. Suitability for Edge Level devices

Edge level devices are restrained by memory and processing power limitations. Neural Networks models with low trainable parameter count and model weights memory size are optimum for these cases. Table III demonstrates the balance posed by proposed model, in terms of mean accuracy and trainable parameters. To obtain similar performance as the proposed model, CNN++ demands 40 times more trainable parameters, which is not feasible for edge level classifier.

TABLE I
NETWORK ARCHITECTURE

Module	Type	Filter No.	Kernel	Output
EEG Net Feature Extractor	Input			(1,C,T)
	Conv2D	8	(1,32)	(8,C,T)
	BatchNorm			(8,C,T)
	DepthWise Conv2D	16	(C,1)	(16,1,T)
	Batch Norm			
	ELU Activation			
	Average Pool 2D			(16,1,T//8)
	Dropout			
	Separable Conv 2D	16	(1,16)	(16,1,T//8)
	Batch Norm			
	ELU Activation			
	Average Pool 2D			(16,1,T//64)
	Dropout			
TCN Classifier	TCN Block 1	4	32	12
	TCN Block 2	4	32	12
	Concatination			24
	Dense			24
	SoftMax Activation			4

C = number of EEG channels, T = number of time samples

TABLE II
RESULTS OF THE EXPERIMENTATION

	Accuracy						Proposed Model
	EEGNet	Shallow ConvNet	EEG TCNet	EEGNet with EOG removed	Shallow Conv Net with EOG Removed	EEG TCNet with EOG removed	
Subject 1	84.34	79.51	85.77	83.24	77.3	81.03	79.3
Subject 2	54.06	56.25	65.02	56.03	57.3	68.97	71
Subject 3	87.54	88.89	94.51	85.13	85.54	75.86	86.2
Subject 4	63.59	80.9	64.91	65.3	82.4	70.7	77.6
Subject 5	67.4	57.29	75.36	65.8	55.35	74.14	81.04
Subject 6	54.88	53.9	61.4	55.6	54.48	62.07	60.03
Subject 7	88.8	91.7	87.36	87.33	89.02	86.21	86.2
Subject 8	76.25	81.25	83.76	78.2	83.2	86.2	89.6
Subject 9	74.24	79.2	78.03	75.7	80.1	91.4	93.1
Mean	72.4	74.31	77.35	72.488	73.85	77.4	80.5
Standard Deviation	13.27	14.54	11.57	12.20	14.07	9.54	9.64

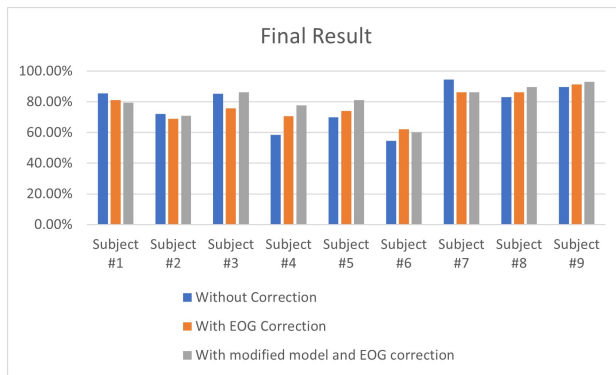


Fig. 3. Subject-Wise Performance Comparison



Fig. 4. Mean and Standard Deviation Comparison

IV. FUTURE PROSPECTS

There are still plenty of opportunity to work on this project. We can further improve the main architecture of the neural network. More direct and light-weight approach for EOG

removal can be used. Building our own dataset will take our work to a great level. There is also scope of expanding number and types of MI tasks.

Our project is a novel work. This knowledge can be inte-

TABLE III
COMPARISON OF TRAINABLE PARAMETERS

	Mean Accuracy	Trainable Parameters	Feature Map [kB]
EEGNet	72.4	2.63k	396
Shallow ConvNet	74.31	47.3k	1013
FBCSP	73.70	261k	50
Riemannian	74.77	50.0k	49
MSFBCNN	75.80	155k	5775
EEG-TCNET	77.34	4.27k	396
CNN++	81.10	240k	499
TPCT	88.87	7.78M	524
Proposed Model	80.5	6.2k	425

grated in physical prototypes, which can be a good assist for physically challenged persons.

V. CONCLUSION

In this paper, we have found some interesting results regarding EEG MI classification using EEG-TCNet. Firstly, we have observed a remarkable reduction of the variation of classification accuracy among test subjects upon EOG artifact removal from EEG data. This result is definitely useful in increasing the efficiency of possible wearable devices running on the EEG-TCNet. Secondly, our modified EEG-TCNET achieves an average accuracy significantly higher than that of currently used EEG MI classification neural networks. At last, we hope that our findings will be helpful for future researchers in the field of EEG MI classification, which will in effect boost the technology of BCI.

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